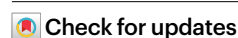


Learning spatiotemporal dynamics with a pretrained generative model

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Reconstructing spatiotemporal dynamics with sparse sensor measurement is a challenging task that is encountered in a wide spectrum of scientific and engineering applications. The problem is particularly challenging when the number or types of sensors (for example, randomly placed) are **extremely sparse**. Existing end-to-end learning models ordinarily do not generalize well to unseen full-field reconstruction of spatiotemporal dynamics, especially in **sparse data regimes** typically seen in real-world applications. To address this challenge, here we propose a **sparse-sensor-assisted score-based generative model (S³GM)** to reconstruct and predict full-field spatiotemporal dynamics on the basis of sparse measurements. Instead of learning directly the mapping between input and output pairs, an **unconditioned generative model is first pretrained**, capturing the joint distribution of a vast group of pretraining data in a self-supervised manner, followed by a sampling process conditioned on unseen sparse measurement. The efficacy of S³GM has been verified on multiple dynamical systems with various synthetic, real-world and laboratory-test datasets (ranging from turbulent flow modelling to weather/climate forecasting). The results demonstrate the sound performance of S³GM in **zero-shot reconstruction** and prediction of spatiotemporal dynamics even with high levels of data sparsity and noise. We find that S³GM exhibits high accuracy, generalizability and robustness when handling different reconstruction tasks.

Understanding and control of dynamical systems, such as fluid dynamics^{1,2}, geophysics^{3,4}, astrophysics⁵ and atmospheric physics⁶, require the information of full spatiotemporal fields (for example, velocity, temperature, pressure and so on). However, in practice, the quantities of interest (QoIs) are **measured under very limited conditions** (for example, the number and/or types of sensors are completely insufficient). In this context, only sparse sensor information is accessible for these complex dynamical systems. Typically, the sparse measurements might include three representative scenarios (Fig. 1b), namely, (1) randomly placed sparse sensors, indicating that the input data are not measured on a regular grid, leading to challenges for spatiotemporal

field reconstruction with unstructured data^{7,8}, (2) incomplete types of randomly placed sparse sensor, where some quantities are inaccessible in a dynamical system with multiple QoIs⁹, and (3) compressed information provided by sensors, for example, in hot-wire velocimetry, which only measures velocity magnitude and loses direction information^{10,11}. These challenges make reconstructing the full-field spatiotemporal dynamics on the basis of heterogeneous sparse measurements critical yet difficult.

Existing data-driven strategies to tackle such a challenge can be broadly categorized into conventional data-driven methods (for example, proper orthogonal decomposition^{4,12–15}, dynamic mode

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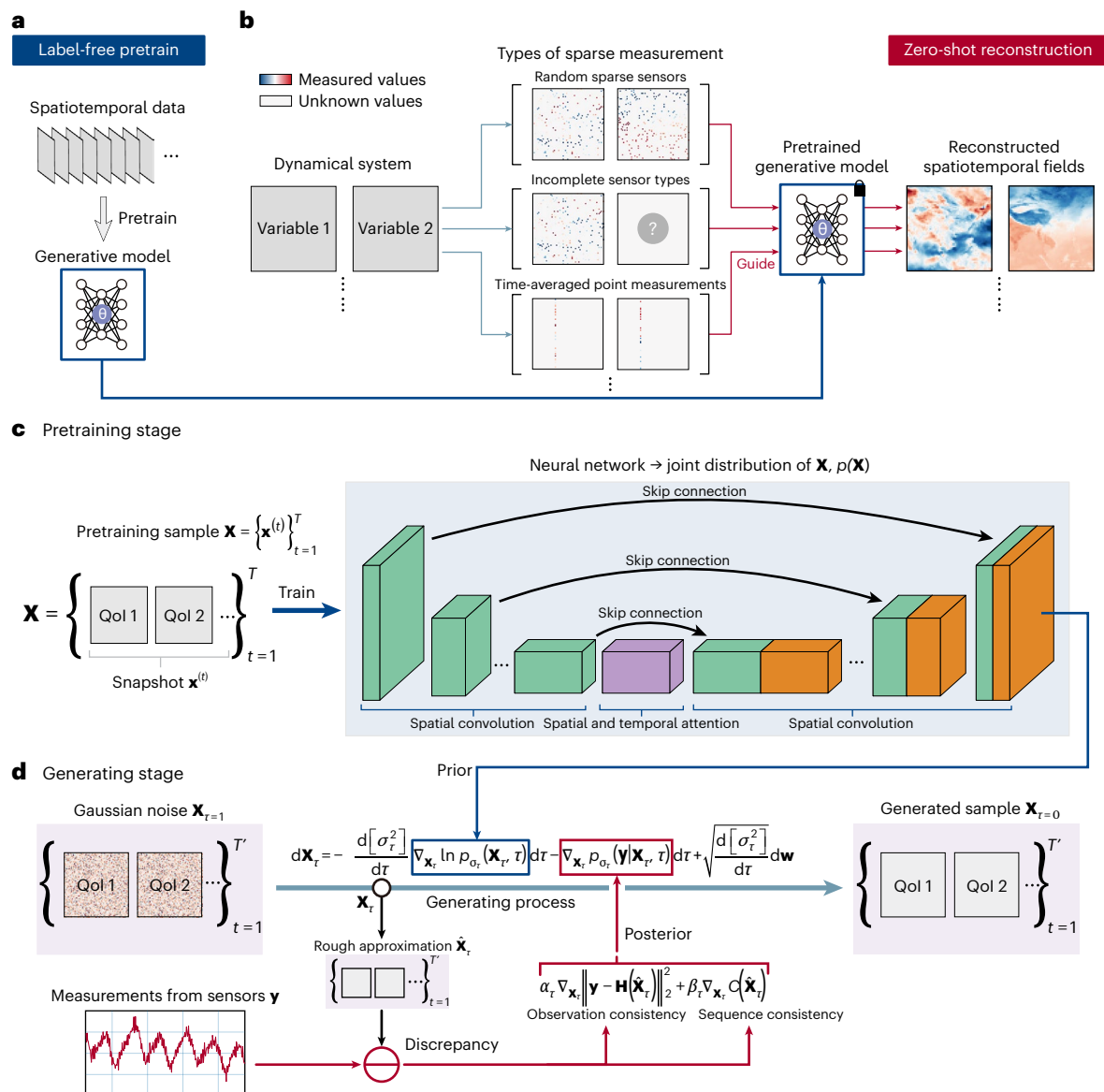


Fig. 1 | Schematic illustration of the proposed S³GM framework.

a, b. Illustration of zero-shot spatiotemporal field reconstruction from different types of sparse measurement with a pretrained GM. **a.** Schematic of self-supervised model pretraining with spatiotemporal data. **b.** Zero-shot spatiotemporal field reconstruction from different types of sparse measurement (for example, the sensor types are complete but randomly sparsely placed; the sensor types are incomplete and randomly sparsely placed; the information provided by sensors is compressed). **c, d.** Demonstration of the two-stage process of the S³GM framework. **c.** In the pretraining stage, the datasets are fed into the

neural network, aiming to characterize the joint distribution $p(\mathbf{X})$ of T snapshots, for example, $\mathbf{X} = \{\mathbf{x}\}_{t=1}^T$, where $\mathbf{x} := \{\text{Qol1}, \text{Qol2}, \dots\}$ represents a snapshot containing all the Qols and T the number of snapshots contained in a training sample. **d.** In the generating stage, with the pretrained model, the sparse sensor information from measurements is incorporated into the sampling process to guide the generation of the full-field dynamics via solving an SDE, where pseudo-time-step τ indicates the generating process and T' the number of snapshots to be generated. Here, $\tau = 1$ denotes the start point of the generating process. When $\tau \rightarrow 0$, the sample is finally generated.

decomposition (DMD)^{16–19}, Galerkin transforms^{20,21} and linear stochastic estimation^{22,23} and neural network (NN)-based methods^{1,2,7,24–33}. The former and its physics-enhanced variants^{34,35}, typically rooted in the linear approximation theory, are very efficient in modelling dynamical systems. However, they suffer from very limited expressiveness and typically require complete sensor measurement for full-field reconstruction. On the other hand, the NN-based methods offer a more powerful alternative for the reconstruction of full-field spatiotemporal dynamics on the basis of sparse data (for example, Voronoi-tessellation-assisted deep learning⁷, randomly seeded super-resolution generative adversarial networks³³). Notably, the majority of NN-based methods establish an end-to-end mapping between input (for example, the sparsely measured field) and target output (for example, full field on

a fixed grid of points) pairs. Hence, for unseen sparse measurements (different sensor types, data sampled from a variant system or under a real-world environment and so on), the trained model cannot make accurate predictions anymore and should be retrained with a new mapping relationship. As a result, these methods only perform well in specific conditions and often suffer from the generalization issue in real-world applications. When adopting the spatiotemporal coordinates as the input for NNs, there exist approaches^{36–38} that allow for continuous reconstruction of the outputs. Therefore, these methods can offer predictions with varying sensor locations or types. However, without the inductive bias introduced by convolution, their capability of capturing complicated spatiotemporal evolution and interaction is very limited. There are also attempts to accelerate the variational data

assimilation by replacing the conventional numerical solver with an operator-based predictive model^{39–42}, which can generalize to different types of observation. However, the non-convex optimization process in variational approaches may suffer from local-optimum issues if the initial value of the optimization is not chosen properly.

One possible solution is to leverage the a priori knowledge of physics principles (for example, governing equations of a particular dynamical system) to constrain the NN training (for example, via a regularized loss function), thus producing a model capable of handling limited data. The seminal development of physics-informed NNs (PINNs)^{9,27,43–47}, for solving differential equation-constrained forward and inverse problems, is conceptually ideal for the reconstruction of full-field spatiotemporal dynamics given sparse measurement data. Nevertheless, once slight changes in the target spatiotemporal field occur (for example, initial and boundary conditions, equation parameters, a similar but different system), a full retraining of the model is required. To alleviate this requirement, attempts have been made to incorporate the equation residue as a penalty constraint into the loss function^{27,47,48}, while neglecting the initial and boundary conditions in the loss to improve the generalization performance. In this way, PINN retraining may be avoided in adapting the model for different applications. However, since the landscape of the PINN loss is often non-convex, it is extremely hard to reach a convergent model with the global optimum when the considered system is highly nonlinear^{49,50}.

Another promising strategy lies in the use of a generative model (GM) that characterizes the distribution of the training data. Instead of modelling the direct mapping between the input and output pairs, the GM learns to generate unseen samples and has the potential to ensure a statistically reasonable output that shares similar patterns with the training data. Recently, the score-based GM (SGM)^{51–55} has exhibited remarkable performance in high-quality generation and provided a flexible and controllable generating process in a wide range of applications without retraining the NN^{53,56–59}. Despite these advantages, employing an SGM for modelling and reconstructing real-world spatiotemporal dynamics presents an underexplored area. Specifically, training an SGM to encode the complex spatiotemporal relations for a dynamical system remains a substantial challenge. Furthermore, when using the SGM to generate spatiotemporal fields, it is challenging to ensure that the pretrained model is able to accommodate diverse inputs from different sparse measurements in real-world configurations. While some pioneering investigations leverage a GM to predict complex dynamics^{25,60–66}, they mainly focus on learning a conditional distribution between input and output pairs in a supervised style. The potential of probabilistic modelling of a data distribution in spatiotemporal field reconstruction has not yet been thoroughly explored.

To this end, a sparse-sensor-assisted SGM (S³GM) framework is proposed in this work to achieve zero-shot spatiotemporal field reconstruction on the basis of various types of sparse measurement. **Here, zero shot refers to the problem set-up where the paired matching between the measurements and their corresponding full-field spatiotemporal dynamics is not provided in the training phase.** Instead of learning directly the mapping between the input and output pairs, an unconditioned SGM is first pretrained by learning the joint distribution of a set of pretraining data in a self-supervised manner, which implicitly encodes the prior physics knowledge of the dynamical system. In the generating stage, with the pretrained model, the sparse sensor information is incorporated into the generating process to guide the generation of the full-field spatiotemporal dynamics. We demonstrate the performance of S³GM by applying it to a set of challenging dynamical systems including synthetic, real-world and laboratory-test data. For a given system, once the GM is pretrained, S³GM can generalize to unseen data and different types of sparse measurement, showing sound performance in zero-shot reconstruction and prediction of spatiotemporal fields even with high levels of data sparsity and noise under conditions beyond the range of the pretraining data. The combination

of prior learning and posterior reconstruction endows S³GM with advantages over representative baseline models and enables S³GM to display high accuracy, generalizability and robustness when handling different reconstruction tasks.

Results

In this section, the framework of the S³GM is introduced, followed by a demonstration of the efficacy of the S³GM on four different dynamical systems including Kuramoto–Sivashinsky dynamics⁶⁷, Kolmogorov turbulent flow^{68–70}, the ERA5 climate observations⁷¹ and laboratory measurements of cylinder flow dynamics. A detailed description of the data acquisition can be found in Supplementary Note 3.

The S³GM method

Spatiotemporal field reconstruction from sparse sensors needs to solve the following inverse problem:

$$\mathbf{y} = \mathbf{H} \left(\{\mathbf{x}^t\}_{t=1}^T \right) + \boldsymbol{\varepsilon}, \quad (1)$$

where $\{\mathbf{x}^t\}_t^T$ denotes the full field across T time steps to be reconstructed and $\mathbf{y}$ represents the sparse measurements from given sensors. $\boldsymbol{\varepsilon}$ follows the Gaussian distribution that represents the overall influence of noise introduced by measurement errors and other uncertain ingredients, and $\mathbf{H}$ denotes the measurement operator, which samples the measurements from a given sequence of states $\{\mathbf{x}^t\}_t^T$ (which can be linear or nonlinear). Typical NN-based models aim to learn the certain inverse mapping of $\mathbf{H}^{-1}$ from noisy measurements $\mathbf{y}$ to the full field $\{\mathbf{x}^t\}_t^T$ in a supervised manner. However, in many practical scenarios, the measurement operator $\mathbf{H}$ may vary, as illustrated in Fig. 1b. For example, the sensors may be randomly placed in space and time, inducing extreme sparsity (Fig. 1b top middle), the sensor types might be incomplete (Fig. 1b middle) or perhaps only statistical quantities (for example, time-averaged quantities) are accessible (Fig. 1b bottom middle). In this context, the trained inverse mapping of $\mathbf{H}^{-1}$ would fail on unseen measurement operators and the time-consuming retraining of the NN would be required.

To address this challenge, a two-stage learning approach (namely, S³GM) is proposed to achieve zero-shot spatiotemporal reconstruction from different types of sparse measurement (Fig. 1a,b). Specifically, in the pretraining stage, we train a GM to characterize the probabilistic distribution of spatiotemporal data (for example, a pretraining set), which implicitly encodes the prior knowledge of the dynamical system into a pretrained GM (Fig. 1a–c). In the generating stage, any type of sparse measurement can be utilized to guide the full-field generation, which satisfies both measurements and prior knowledge of the dynamics stored in the pretrained model (Fig. 1b–d and Extended Data Fig. 2). The details of the methodology and implementation can be found in Methods.

Kuramoto–Sivashinsky dynamics

As a representative model for a spatiotemporally chaotic system, we consider first the Kuramoto–Sivashinsky equation (KSE⁶⁷), which is extensively studied in the context of instability in reaction–diffusion systems and transition of turbulence⁷². The KSE is written as

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} + \mu \frac{\partial^4 u}{\partial x^4} = 0, x \in \Omega, \quad (2a)$$

$$u(0, x) = u_0(x), x \in \Omega, \text{ and } u(t, x) = u_b(t), x \in \partial\Omega, \quad (2b)$$

where u denotes the solution field. t and x refer to the temporal and spatial coordinates, respectively, and parameter μ denotes the viscosity. Given a proper initial condition u_0 and boundary condition u_b , the solution within a specified physical domain can be determined. Due to the chaotic instinct of the system, different settings of initial conditions

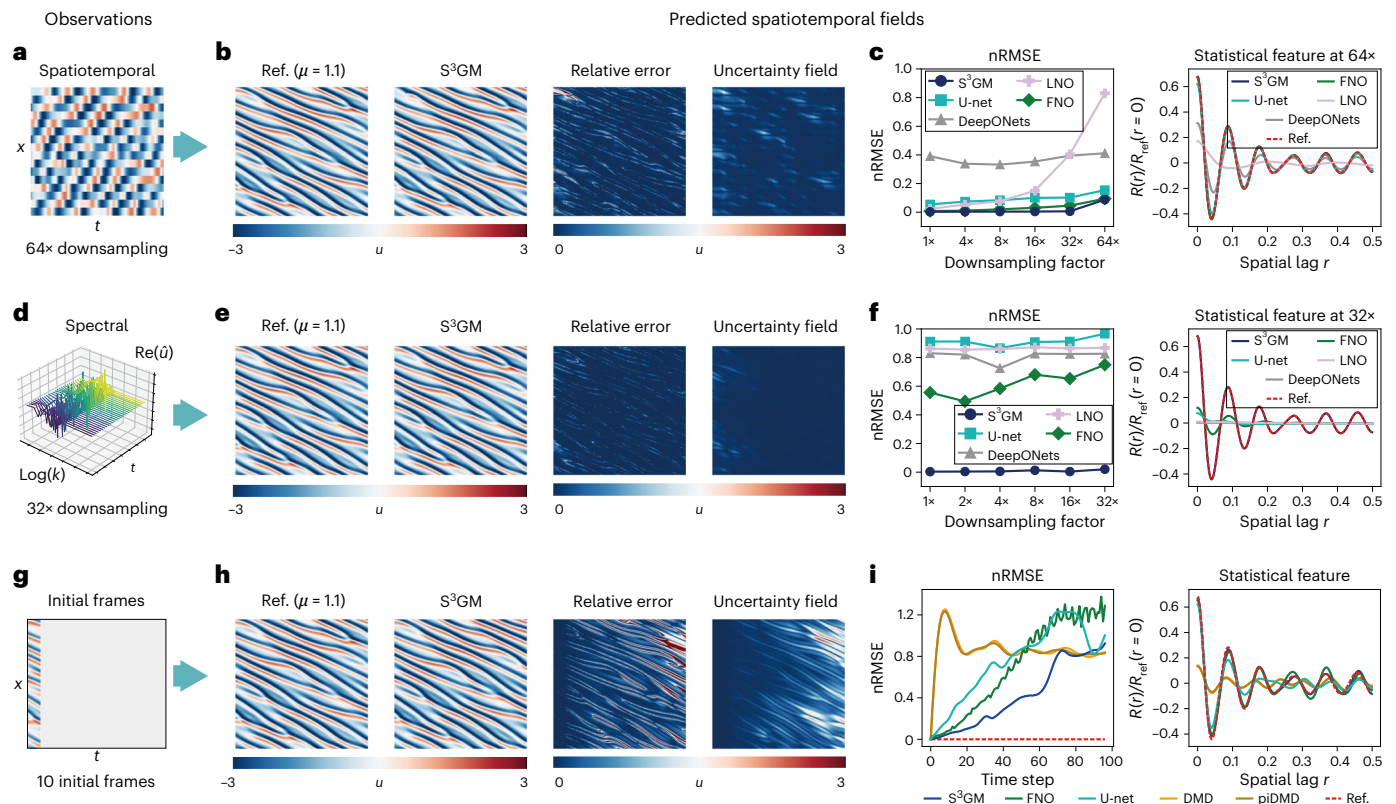


Fig. 2 | Reconstruction of KSE solutions under different types of sparse measurement. **a**, Sparse measurements in the spatiotemporal space. The spatial downsampling rate is 64, and the number of discretized time steps is 100. **b**, Reconstructed solution in the physical space using sparse measurements. From left to right, the panels represent the high-fidelity simulation (denoted as Ref.), the S³GM prediction, the relative error field and the uncertainty field. **c**, Prediction error and statistical features (spatial two-point correlation) of S³GM and baseline methods (note that the baseline models are individually trained on each of the downsampled datasets). **d**, Sparse observations in the spectral space, which are chosen to be the real part of Fourier features from the spatially

downsampled solution with a factor of 32. **e**, Reconstructed solution in the physical space using measurements in the spectral space. **f**, Relative error and statistical features of S³GM and baseline methods. **g**, Initial frames for forward forecasting. **h**, Predicted solution in the physical space on the basis of the initial frames. **i**, Prediction error and statistical features of S³GM and baseline methods (trained to predict the next snapshot with the previous ten snapshots as input). All the uncertainty fields in this figure are the s.d. fields, calculated from five different predictions given the same observation (multiple predictions are acquired by repeating the generation process with different random seeds).

and viscosity cause deviated spatiotemporal evolution of solutions (Supplementary Fig. 1). Therefore, to obtain prior knowledge of this dynamical system, we characterize the joint distribution of QoIs in this system (that is, u and μ) by pretraining from high-fidelity simulation data (see Methods and Supplementary Note 3 for detailed information), where we employ periodic boundary conditions and the samples from a zero-mean one-dimensional Gaussian process as the initial conditions.

To evaluate the performance of S³GM, the full-field dynamics reconstructed from limited sparse observations are examined (Fig. 2). Here we consider three different types of observation: spatiotemporal measurements (Fig. 2a), spectral measurements (Fig. 2d) and initial frames (Fig. 2g). Overall, S³GM is able to reconstruct the complicated spatiotemporal patterns of the solution as well as the parameter μ . Specifically, the solution for testing starts from an unseen initial condition and evolves for much longer than the training sample. In Fig. 2a, the sparse measurements are sampled from high-fidelity numerical simulations (with a spatial downsampling rate of 64). It can be seen from Fig. 2b that the proposed framework can accurately reconstruct the original solution given sparse spatiotemporal measurements, achieving relatively low error and uncertainty averaged on multiple samples. In Fig. 2c, compared with other end-to-end baselines (Fourier neural operator (FNO)³⁹, U-Net⁷³, Laplace neural operator (LNO)⁷⁴ and DeepONets⁷⁵, Methods) individually trained on each of the downsampled datasets (1×, 4×, 8×, 16×, 32× and 64×), S³GM can still attain better performance in the sense of the normalized relative mean square error (nRMSE) (see the

definition in Methods) and statistical feature (that is, spatial two-point correlation, Methods) without retraining our model. The generalization of S³GM can be further illustrated by forcing the un-retrained baselines to reconstruct from observations with unseen downsampling factors (Supplementary Fig. 12). Furthermore, S³GM can simultaneously recover the corresponding parameter μ during reconstruction (Supplementary Fig. 16). Note that the increase of the data sparsity (that is, increasing the downsampling rate) may lead to a deteriorated reconstruction performance (Fig. 2c and Supplementary Figs. 10 and 11). The S³GM framework can also promote flexible field reconstruction on the basis of different types of measurement, such as measurement in the spectral space (Fig. 2d). We use the real part of Fourier features of the solution spatially downsampled from the original solution (with the rate of 32) for reconstruction. The results demonstrate a similar performance, as shown in Fig. 2e, showing low relative errors and uncertainty. In this case, the baselines (FNO, U-Net, DeepONets and LNO) are trained to learn the reconstruction from spectral observation at each downsampling rate (1×, 2×, 4×, 8×, 16× and 32×). Since the mapping between the spectral space and the spatiotemporal space is intricate, the end-to-end baseline models can hardly predict the spatiotemporal field from measurements in the spectral space, while S³GM performs well under different degrees of observation sparsity (Fig. 2f and Supplementary Fig. 13). Here, although the observation is sampled in the spectral space, the reconstructed response still lies in the spatiotemporal physical space. Hence, the error is calculated in the physical space, as reported in Fig. 2e,f. Moreover, we

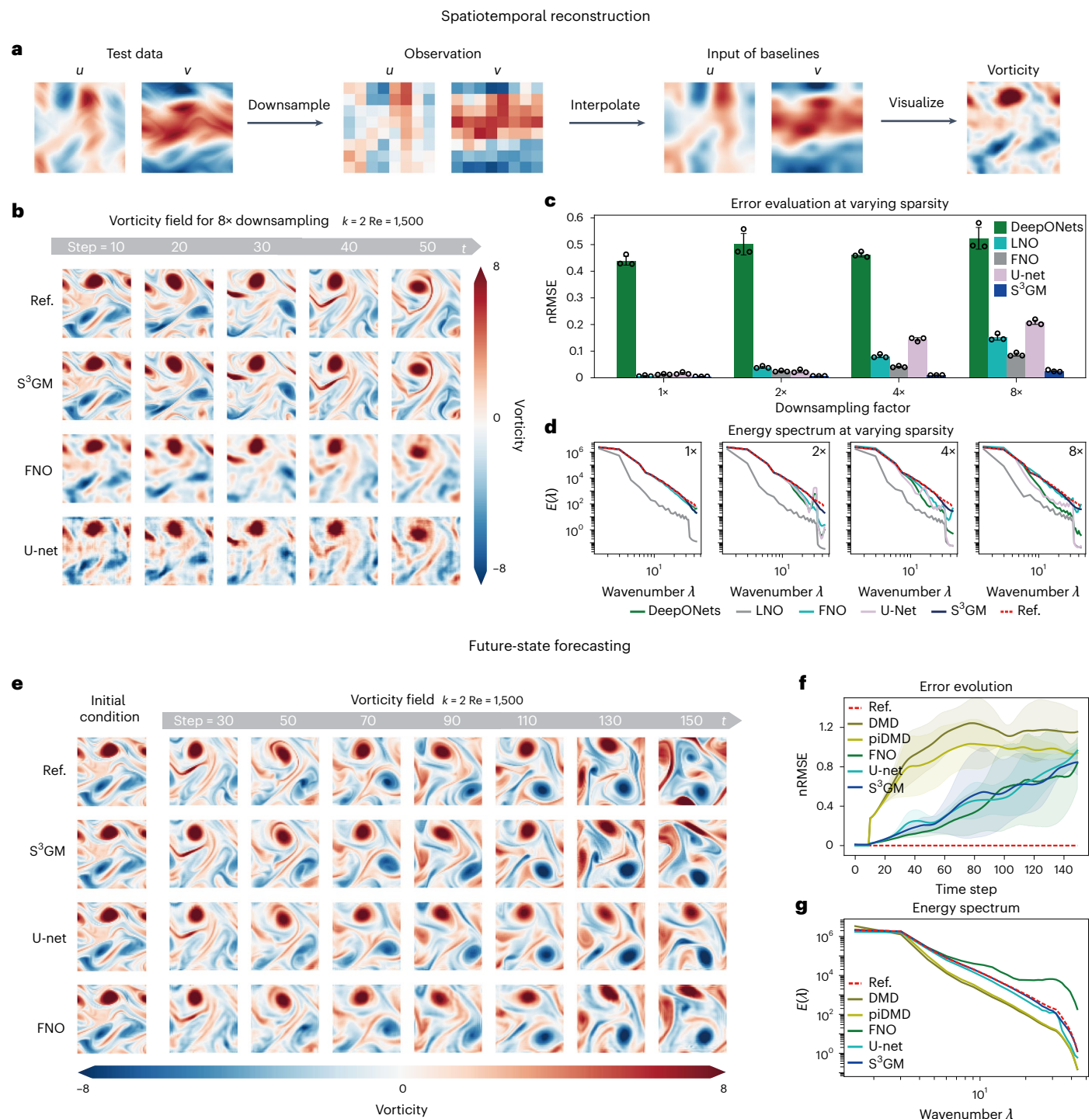


Fig. 3 | Reconstruction of Kolmogorov turbulent flow at a high Reynolds number that is not seen in training data. a–d, Quantitative results for reconstructing a long sequence of turbulent flow fields from sparse spatiotemporal observations. **a**, Sparse observations sampled from high-fidelity numerical simulations (denoted by Ref.). **b**, Comparison between the fields reconstructed from 8× downsampling at $k = 2$, $Re = 1,500$. Here, the baseline models are trained on the 8× super-resolution task. **c**, The prediction errors of different methods under varying sparsity (the baselines are separately trained on each of the downsampling factors). The error bars represent the mean values $\pm$ s.d., calculated from the method's predictions across three different

solutions with varying initial conditions. **d**, Kinetic energy spectrum of the predicted and simulated flow fields at different degrees of sparsity (1×, 2×, 4× and 8× downsampling). **e–g**, Results for future-state forecasting, that is, using S^3GM to make future predictions of the flow fields given several initial frames of the flow field. **e**, Flow fields predicted using U-Net, FNO and S^3GM . Here, we visualize the vorticity fields instead of the original velocity fields. **f**, Error accumulation of different models under $k = 2$ at $Re = 1,500$. The error bands present the mean values $\pm$ s.d. of the predictions that start from three different initial conditions. **g**, Comparison of statistical features for the turbulent flow in the context of the kinetic energy spectrum.

also show the numerical stability of S^3GM for response forecasting. Given several initial frames of the solution trajectory, S^3GM can make stable predictions for a long period (Fig. 2h). We further compare our method with other neural baseline models (FNO and U-Net, Methods and Supplementary Note 2) for future-state forecasting. The results show that S^3GM produces more accurate and stable predictions, outperforming these baseline approaches (Fig. 2i and Supplementary Figs. 14 and 15). We also compare our model with linear methods such as DMD¹⁶ and physics-informed DMD (piDMD)³⁴ (Methods and Supplementary Note 7), with the test results on the KSE dataset presented in Supplementary Fig. 9. An animation of the above predictions is further shown in Supplementary Video 1.

In addition, to comprehensively illustrate the generalizability of the S^3GM framework, we also conducted experiments on one-dimensional cases of the Korteweg–de Vries⁷⁶ system (Supplementary Note 4 and Supplementary Fig. 5) and Burgers' equation (Supplementary Note 6 and Supplementary Fig. 8) to further validate the proposed method.

Kolmogorov turbulent flow

To further demonstrate the capability of S^3GM in modelling complex dynamical systems, we test a more complicated system, that is, Kolmogorov turbulent flow^{68–70}. Governed by the Navier–Stokes equation, the Kolmogorov flow dynamics are highly nonlinear and chaotic combined with the co-existing multiscale structures, resulting in significant challenges in full-field reconstruction and prediction. Moreover, the behaviour and characteristics of such a dynamical system can be distinct when changing the Reynolds number Re or the external force $\mathbf{f}$ (see Supplementary Fig. 2 for example). Characterizing the turbulence under diverse source terms and Reynolds numbers introduces great demands for the high capacity and generalizability of the model.

In this case, we generate the training data using typical configurations suggested in most two-dimensional turbulence simulations that use random Gaussian fields constrained to a specific energy spectrum as initial conditions^{68,77}. After attaining statistical steady state, we collect the training data. The prior distribution learned from the data jointly captures the interactions between all the QoIs (that is, velocity $\mathbf{u}$, $\mathbf{f}$ and Re). After pretraining the model on high-fidelity simulation data with k , $Re \in \{2, 3, \dots, 8\} \times [100, 1,050]$ (where k is a parameter that controls the external force), the model is evaluated on a higher Reynolds number ($Re = 1,500$) to check its generalizability and numerical stability (Fig. 3). As demonstrated in Fig. 3a,b, the long-time sequence of the velocity fields can be accurately reconstructed from sparse (for example, $8\times$ downsampled along each spatial dimension) measurements, even without knowing the external force and the Reynolds number. This proves the good generalizability and robustness of S^3GM . Furthermore, S^3GM remains advantageous when compared with other end-to-end neural models (for example, FNO, U-Net, LNO and DeepONets, Methods) that are specifically trained for the $8\times$ super-resolution task (Fig. 3b).

To further illustrate the performance of S^3GM , the discrepancy between the reference and the model's prediction is quantified under different levels of data sparsity with $1\times$, $2\times$, $4\times$ and $8\times$ spatial down-sampling, respectively (see Fig. 3c,d). For pointwise accuracy, the nRMSE is evaluated between the predicted solution and the reference with different sparsity. We observe that S^3GM produces relatively low errors in the velocity field even for larger Reynolds numbers excluded from the training data (Fig. 3c). On the other hand, the baseline models (FNO, U-Net, DeepONets and LNO) show larger reconstruction errors even when specifically trained on each of the downsampled datasets (Fig. 3c). Given the fact that the pointwise accuracy cannot fully represent the model's performance (as shown in Extended Data Fig. 1), the consistency in the spectral space (namely, the kinetic energy spectrum) is compared. The performance of S^3GM is robust to the change of sparsity of the observations (Fig. 3d). It is found that, compared with

end-to-end baselines, S^3GM is capable of accurately recovering the spectral features.

Apart from the capability of handling spatiotemporal observations, S^3GM also exhibits notable performance in forward forecasting (Fig. 3e–g). We start from ten initial frames of the velocity fields to predict the flow evolution for 150 future steps (Fig. 3e). Given the initial conditions and parameters that are not seen in the pretraining stage, S^3GM can correctly predict the flow evolution at a long-term horizon as shown in Fig. 3f.

Furthermore, compared with other data-driven models—for example, U-Net and FNO (Methods), which are particularly designed and trained for this task— S^3GM provides competitive results in the context of pointwise accuracy (Fig. 3f). Notably, although the FNO and U-Net models seem to achieve similar performances in terms of the nRMSE metric, they might produce distinct errors and unphysical patterns (Fig. 3e), indicating numerical instability during forecasting. Consequently, the future-state forecasting by the baseline models is less reliable. This can be further confirmed by the energy spectrum, where S^3GM is more consistent with the statistical features of the turbulence, with better generalizability over the Reynolds number beyond (Fig. 3g). In addition, the linear methods such as DMD and piDMD (Methods and Supplementary Note 7) fail to capture the dynamics of the chaotic turbulence due to their simple linear assumption (Supplementary Fig. 9). Furthermore, due to the stochastic nature of the generative modelling, S^3GM is capable of providing uncertainty quantification for the predictions (Supplementary Figs. 19 and 20). We also investigate our model's performance of response forecasting with imperfect initial conditions. It turns out that our method shows even better performance when compared with the baseline models on both randomly located observations (Supplementary Figs. 17 and 21) and regularly located observations (Supplementary Figs. 18 and 22). More details can be found in Supplementary Note 8. An animation of the predicted Kolmogorov flows is shown in Supplementary Video 2. In addition to the incompressible flow depicted in Fig. 3, we also validate S^3GM on compressible fluid dynamics, with detailed results and discussion reported in Supplementary Note 5 and Supplementary Figs. 6 and 7.

ERA5 climate observations

The monitoring and prediction of global climate evolution rely on the appropriate reconstruction of multi-variables from sparse and noisy observations^{3,4}. In this example, the training and testing data are collected from the ERA5 reanalysis⁷¹, which is a large dataset containing hundreds of climate-related quantities ranging from the year 1940 till now. Here, we choose a subset of ERA5 that includes four frequently used QoIs: the u component of wind at 10 m height, the v component of wind at 10 m height, temperature at 2 m height and pressure on the earth's surface (denoted as u , v , Temp and P , respectively). We train our model on historical data from 1979 to 2022 to characterize the joint distribution of the four QoIs, while testing it using the data in 2023 (see Supplementary Note 3 for details).

To illustrate the capability of S^3GM , we merely provide the model with random spatiotemporal points, which are only 1% of the testing data (Fig. 4a). We further add an additional 10% of Gaussian noise to the sparse observations to examine the robustness of S^3GM . Moreover, the quantities Temp and P are totally masked, for example, unobserved, in the generating stage (Fig. 4a). Even under such adverse conditions, S^3GM can still generate comparable reconstruction results of all QoIs covering a long period of 100 snapshots (that is, 100 d in our configuration, as shown in Fig. 4b,c). It is noted that the reconstructed fields depicted in Fig. 4b,c show the worst case, since only 1% of the noisy data are available. To further quantify the prediction errors, we calculate the nRMSE and the correlation (a metric that evaluates the pointwise accuracy as shown in Methods) between the predicted fields and the ground truth under different numbers of sensors (Fig. 4d,e). As the amount of observed information increases, the total prediction errors

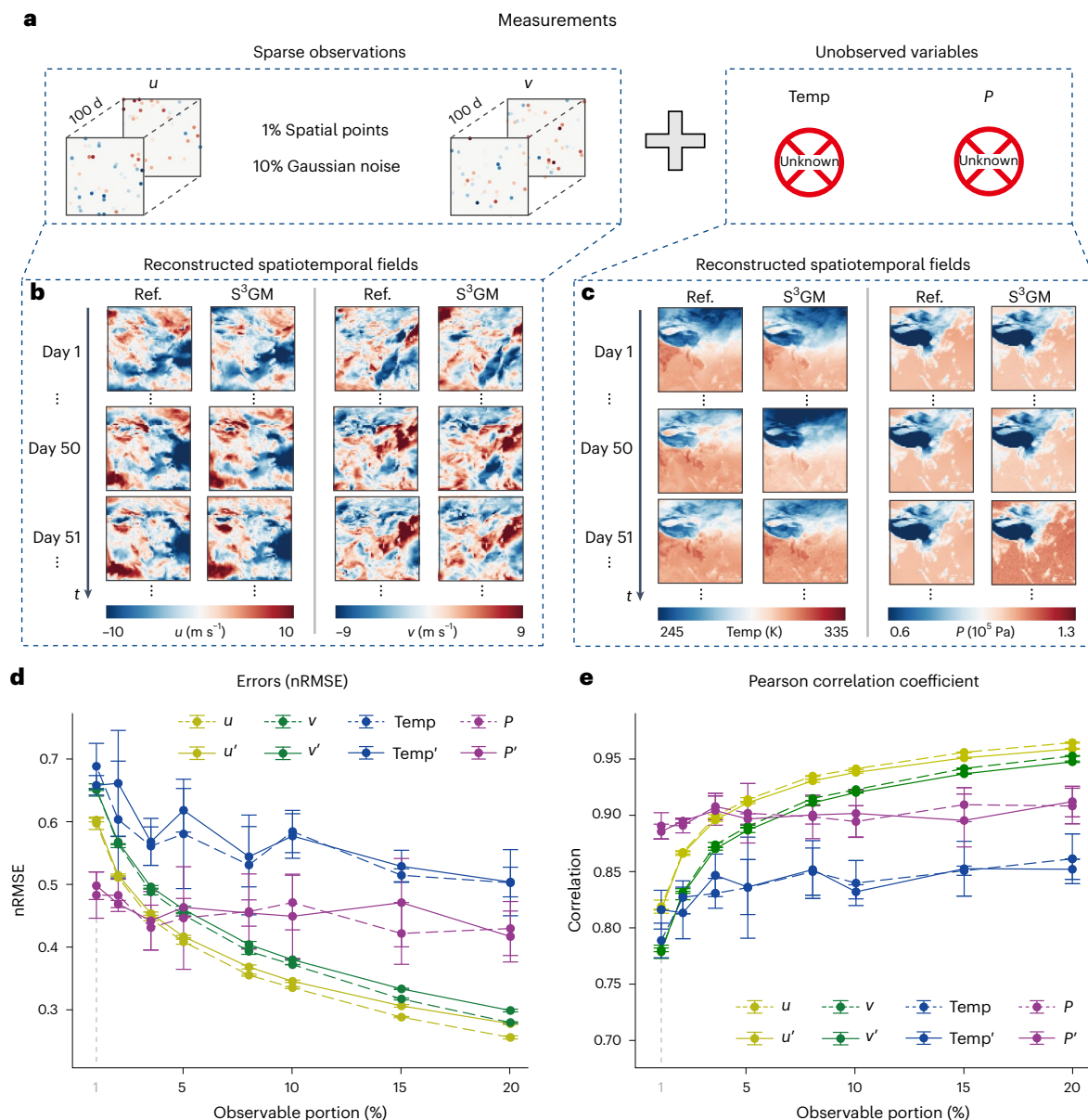


Fig. 4 | Reconstruction of the climate data in 2023 from incomplete and noisy observations, in which S³GM is pretrained using the historical data from 1979 to 2022. **a, Among the four quantities (that is, u , v , Temp, P), only 1% of spatial points of noisy u and v are set to be observable, while Temp and P are not given. Temp, temperature. **b**, Comparison between the ground-truth data (denoted as Ref.) and the reconstructed spatiotemporal fields (denoted as S³GM) on the observed quantities u and v . **c**, Comparison between ground-truth data and the reconstructed spatiotemporal fields on the unobserved quantities Temp and P . **d**, Prediction errors of u , v , Temp and P as the number of observable points increases. The solid line and the superscript ' represent the results predicted**

from noisy observations, while the dashed line represents the results predicted from clean observations. **e**, Correlation between the ground-truth data and S³GM predictions as the number of observable points increases. The error bar in this figure is the deviation for three different predictions generated with different random seeds for the same observation. In **d** and **e**, data are presented as mean $\pm$ s.d., calculated from three predictions given the same observation (by varying the random seeds to obtain different predictions generated by our model). The grey vertical dashed line indicates the smallest observable portion (1%), which signifies the starting point of the curves in **d** and **e**.

quickly decrease, with the correlation reaching close to 1. The results show that adding noise in the sparse observations only slightly reduces the model performance, demonstrating the satisfactory robustness of S³GM. The uncertainty fields of the predictions can be found in Supplementary Fig. 24. An animation of the predicted ERA5 dataset is shown in Supplementary Video 3.

Laboratory test of cylinder flow dynamics

We further consider a classical phenomenon—flow past a circular cylinder (for simplicity, we use the term ‘cylinder flow’ throughout this Article),

which can be described by the Navier–Stokes equation. With a prior distribution of the cylinder flow pretrained using simulated data containing QoIs of u and Re (Methods), the performance of S³GM is examined by reconstructing the velocity field u utilizing a set of laboratory experimental data. The instruments used and schematics of the experimental system are represented in Fig. 5a and Extended Data Fig. 3, where we acquire the velocity vector field using the particle image velocimetry (PIV) technique. The sparse measurements are manually downsampled from the collected PIV data to mimic the exact measurements from a hot wire, which is an intrusive device that can only measure pointwise velocity amplitude.

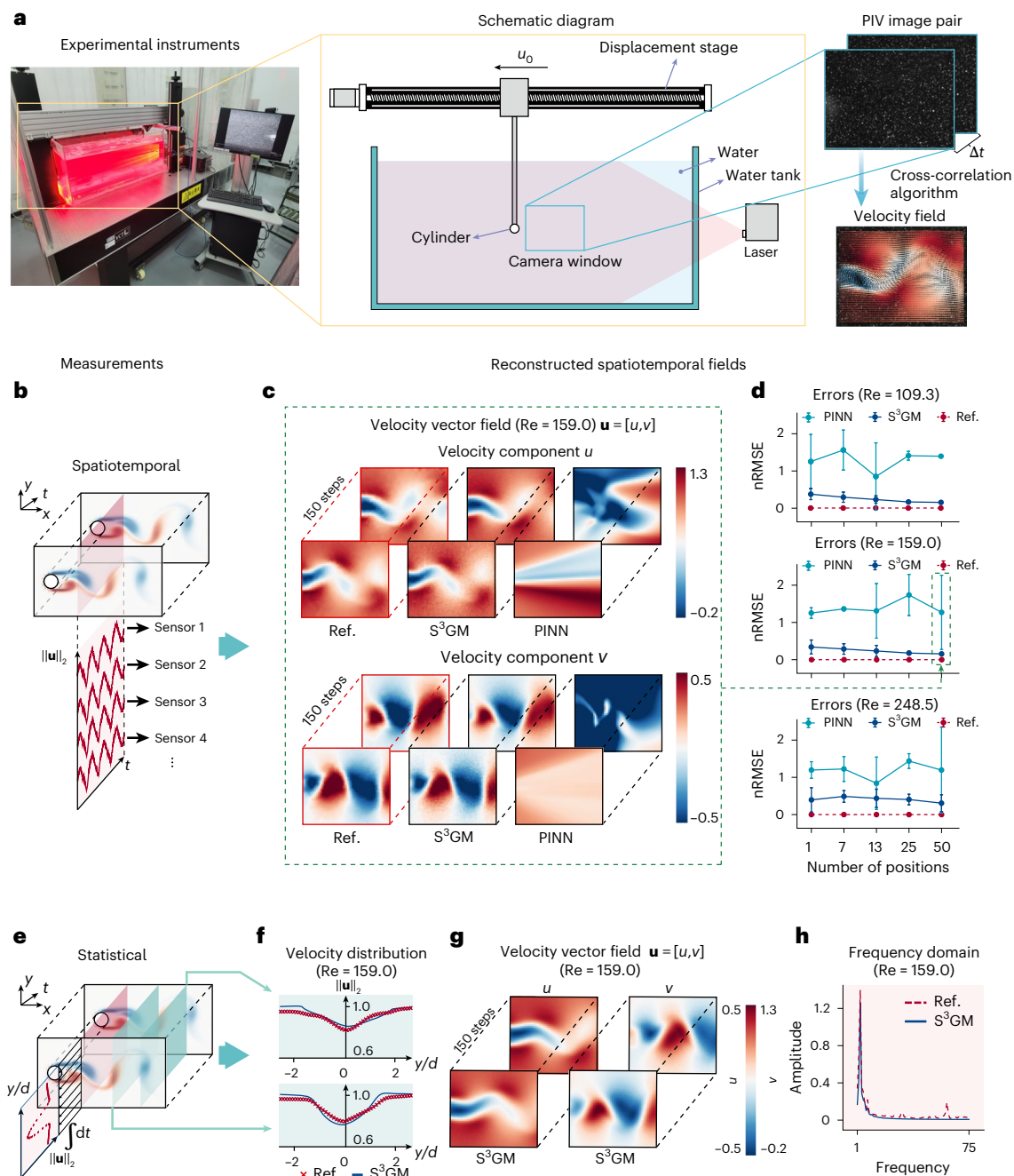


Fig. 5 | Reconstruction of the cylinder flow from different types of real-world experimental datum. **a**, Experimental instruments used for conducting the PIV measurements of the cylinder flow, including a picture of the whole experimental set-up (left), a schematic diagram of the experimental system (middle), an instance of a PIV image pair captured by the image sensor (top right) and the processed velocity vector field calculated using a cross-correlation algorithm (bottom right). **b–d**, The predicted velocity field from spatiotemporally distributed observations. **b**, Illustration of the measurements located at a downstream cross-section. Each of the measured signals is a long-time sequence representing the amplitude of the velocity. **c**, A sample of the velocity vector fields measured using PIV (left, denoted as Ref.), predicted by the S^3GM (middle) and predicted by a PINN (right). **d**, The prediction error

(evaluated using nRMSE) on this task at varying Reynolds numbers and observed spatial locations. Data are presented as mean $\pm$ s.d., calculated from multiple predictions given the same observation. For the S^3GM , ten different predictions are acquired by repeating the generating process with different random seeds. For the PINN, three different predictions are acquired by varying the initial parameters of the NN. **e–h**, The predicted velocity field from statistical quantities. **e**, Illustration of the measurements located at the same cross-section, where each of the locations contains only a time-averaged velocity amplitude and d denotes the diameter of the circular cylinder. **f**, Time-averaged velocity distribution at two unseen cross-sections. **g**, Velocity vector fields reconstructed using the S^3GM . **h**, Comparison of the Fourier features between the experimental data and the predicted velocity field.

As demonstrated in Fig. 5b,e, the sparsely collected data can be roughly divided into two types. The first type is spatiotemporally distributed (Fig. 5b), while the second type is the statistical value that is usually expressed in the form of integrals over physical quantities

(for example, the time-averaged velocity distribution shown in Fig. 5e). The data for the first type of observation are spatially sampled at a single downstream cross-section, where a long sequence of undirected velocity amplitude is measured at each of the locations (Fig. 5b).

The velocity field predicted using the S³GM is compared with that predicted using the PINN at $Re = 159$, as shown in Fig. 5c. When compared with the exact velocity field measured using PIV (denoted as Ref. in Fig. 5c), the sequence generated using the S³GM can capture the important flow features, while the PINN fails to recover the flow pattern. Moreover, the S³GM produces lower prediction errors than does the PINN (Fig. 5d). To further illustrate the performance of the S³GM under different conditions, the nRMSE metric at three different Reynolds numbers (109.3, 159.0 and 248.5) is quantified and the influence of the number of sensors on the cylinder flow reconstruction is also examined (Fig. 5d). The results show that the S³GM statistically achieves higher accuracy in these conditions when compared with the PINN (Fig. 5d).

To test the second type of measurement, we choose the same measurements of undirected velocity amplitude sampled from a downstream cross-section, where the only change is the averaged temporal information of signals (Fig. 5e). This time-averaged observation is ubiquitous in practice^{57–59} because some of the intrusive devices (such as a Pitot probe or hot-wire velocimetry) can hinder the simultaneous arrangement of multiple devices, making it impossible to obtain synchronized signals at multiple spatial positions in a single experiment. In this situation, the only spatial information we can gather is the time-averaged velocity, which reflects the statistical features of the dynamics while the information is highly compressed when compared with the first type of observation. It is seen that S³GM can predict the transient dynamics that satisfy the observed statistical features (Fig. 5g). We further compare the time-averaged velocity field (Fig. 5f) and the Fourier features (Fig. 5h) between S³GM and the PIV measurements. The velocity distribution predicted by S³GM exhibits a high correlation with the experimental results at unseen cross-sections, as shown in Fig. 5f. On the other hand, the Fourier features predicted by S³GM in the frequency domain match the measurements well, and hence the frequency of vortex shedding is accurately recovered (Fig. 5h). The uncertainty quantification of the predictions using S³GM can be found in Supplementary Fig. 25. An animation of the predicted cylinder flow is shown in Supplementary Video 4. Notably, in such a system with explicit periodic spatiotemporal patterns, linear methods (for example, DMD and piDMD) possess excellent capacity to efficiently characterize the dynamics (the comparison of our model with these linear methods is discussed in Supplementary Note 7 and Supplementary Fig. 9). However, for much more complicated systems involving chaotic behaviours or wave propagation (for example, Kolmogorov flow and KSE), S³GM significantly outperforms these linear methods. As a general approach, S³GM also shows remarkable capability in handling observations under real laboratory conditions.

Discussion

This Article introduces an SGM framework, namely, S³GM, for the zero-shot reconstruction of full-field spatiotemporal dynamics from various types of sparse measurement. This effort opens a new pathway toward the use of pretrained GMs for modelling and predicting spatiotemporal dynamics in small data regimes. Instead of directly learning the mapping between input and output pairs, S³GM consists of two-stage modelling processes. In the pretraining stage, an unconditioned SGM is trained on the basis of a set of pretraining data to acquire comprehensive information on the considered dynamical system in a self-supervised manner. In the generating stage, with the pretrained GM and the designed sampling path, the missing spatiotemporal dynamics (for example, the full-field dynamics) can be effectively reconstructed from various types of sparse and noisy measurement. The combination of prior learning and posterior reconstruction allows the proposed S³GM to possess strong flexibility and robustness when handling different reconstruction tasks. This is because learning the joint distribution of the pretraining data implicitly encodes the key physics of the dynamical system, thereby reflecting the prior knowledge of the system and enhancing the generalizability of S³GM.

The performance of S³GM has been extensively verified for learning spatiotemporal dynamics from sparse measurements via multiple synthetic, real-world and laboratory experimental datasets, including Kuramoto–Sivashinsky dynamics, chaotic Kolmogorov flow, real-world ERA5 climate observations and a laboratory experiment on flow past a circular cylinder. The results show sound accuracy, generalizability and robustness of the proposed S³GM in reconstruction and forecasting of full-field spatiotemporal dynamics even with high levels of data sparsity and noise. While S³GM does not explicitly encode prior knowledge of physics and/or physical constraints, satisfactory matches with the underlying physics are found in the analysis of predicted quantities (for example, the energy spectrum). This is attributed to the advantage of S³GM in implicitly encoding the key features of the dynamics by learning the joint distribution of the pretraining data. Comparison with several baseline methods (for example, PINN, FNO, U-Net, DeepONets, LNO, DMD and piDMD) indicates that the proposed S³GM framework has notable generalizability and robustness for different types of sparse and noisy measurement.

While the S³GM framework shows promise in zero-shot reconstruction and forecasting of complex spatiotemporal dynamics, it suffers from relatively high computational costs, especially when the high-dimensional pretraining data involve long-term evolution (a list of computational costs for each model is shown in Supplementary Table 5, with the corresponding runtime environment stated in Supplementary Note 12). Moreover, as an NN-based data-driven method, S³GM requires rich pretraining data (for example, high fidelity) to perform well in the generating stage, especially for the more challenging task of future-state forecasting (the training data usage for each case is listed in Supplementary Table 4). Nevertheless, this limitation is expected to be alleviated by either adding explicit physical constraints during pretraining or learning the latent space with reduced dimensions⁷⁸. In addition, once the GM is pretrained, S³GM can generalize to unseen data and different types of sparse measurement for a given dynamical system without retraining. However, the model's generalizability across different types of dynamical system still faces potential risks and challenges. This limitation could be resolved by designing a model that characterizes the distribution of local features of the spatiotemporal field instead of the global field in the pretraining stage. Finally, the generating stage, resorting to numerically solving a **stochastic differential equation (SDE)**, results in a low generation speed. This issue can be improved by using a more advanced sampling method^{79,80} or distilling the pretrained model⁸¹ to directly predict the denoised fields. We aim to systematically address these potential limitations in future work.

Methods

The S³GM

In the following, we provide a detailed explanation of the two-stage learning paradigm (that is, pretraining stage and generating stage) employed by S³GM.

Overview of the S³GM. In the pretraining stage, the proposed S³GM first approximates the joint distribution of T snapshots $\mathbf{X} = \{\mathbf{x}^t\}_{t=1}^T \in \mathbb{R}^d$ via an SGM^{53,55} (Fig. 1c), where $\mathbf{x}$ contains the QoIs that depend on the studied system (for example, for Kolmogorov flow, $\mathbf{x}^t = [\mathbf{u}, Re, k]^T$). After pretraining, we further estimate the conditioned generative probability $p(\mathbf{X}|\mathbf{y})$ with the observation $\mathbf{y} = \mathbf{H}(\mathbf{X}) + \mathbf{e}$, as shown in equation (1), for generating the desired contents under certain observations $\mathbf{y}$ (Fig. 1d).

Model pretraining for estimating the joint distribution. The SGM^{53,55} is employed to learn the joint distribution $p(\mathbf{X})$ from a discretized dataset containing instances of $\mathbf{X}$. As shown in previous work^{53,55}, SGM does not directly fit the data distribution $p(\mathbf{X})$ itself but approximates the score of the distribution $\mathbf{s}(\mathbf{X})$, defined as the gradient of the log-likelihood with regard to the sample, given by

$$\mathbf{s}(\mathbf{X}) = \nabla_{\mathbf{X}} \ln p(\mathbf{X}). \quad (3)$$

The score of the distribution can be tightly approached⁵³ by introducing different levels of Gaussian noise σ_τ , $\tau \in [0, 1]$ into the following score-matching objective:

$$J(\theta) = \mathbb{E}_\tau \left[\mathbb{E}_{p(\mathbf{X})} \left[\mathbb{E}_{p_{\sigma_\tau}(\mathbf{X}_\tau|\mathbf{X})} \left[\|\mathbf{s}_\theta(\mathbf{X}_\tau, \tau) - \nabla_{\mathbf{X}_\tau} \ln p_{\sigma_\tau}(\mathbf{X}_\tau|\mathbf{X})\|_2^2 \right] \right] \right], \quad (4)$$

where $\sigma_\tau > 0$ is the scheduled s.d., $p_{\sigma_\tau}(\mathbf{X}_\tau|\mathbf{X}) = \mathcal{N}(\mathbf{X}_\tau; \mathbf{X}, \sigma_\tau^2 \mathbf{I}_d)$ represents a series of Gaussian distributions with data perturbed by different levels of noise, and the score approximating NN $\mathbf{s}_\theta(\mathbf{X}_\tau, \tau)$ evaluates the scores at each perturbing stage τ .

Given a training sample $\mathbf{X} \sim p(\mathbf{X})$, we perturb it at step τ as $\mathbf{X}_\tau = \mathbf{X} + \sigma_\tau \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ denotes a random variable sampled from a standard Gaussian distribution. Considering that $p_{\sigma_\tau}(\mathbf{X}_\tau|\mathbf{X}) = \mathcal{N}(\mathbf{X}_\tau; \mathbf{X}, \sigma_\tau^2 \mathbf{I}_d)$ is a Gaussian distribution, we can easily obtain its score $\nabla_{\mathbf{X}_\tau} \ln p_{\sigma_\tau}(\mathbf{X}_\tau|\mathbf{X}) = \boldsymbol{\epsilon}/\sigma_\tau$ using the reparameterization trick⁸². Substituting the expression of $\mathbf{X}_\tau$ and $\nabla_{\mathbf{X}_\tau} \ln p_{\sigma_\tau}(\mathbf{X}_\tau|\mathbf{X})$ into equation (4), the NN $\mathbf{s}_\theta(\mathbf{X}_\tau, \tau)$ can be well trained with a properly chosen architecture and adequate data supplies (a description of the training data requirements can be found in Supplementary Note 9 and Supplementary Fig. 26). Here, to better characterize the temporal dynamics, we choose an architecture called Video U-Net^{83,84} to model the sequential correlations among snapshots. The details of the training process are provided in Supplementary Note 1 and Supplementary Table 1. Our method can also be extended to the training data on non-Cartesian grids using graph NNs⁸⁵ or sparse-convolution-based NNs^{86,87} (see Supplementary Note 13 and Supplementary Fig. 29 for more details).

Sampling long sequences from pretrained score. We further consider the controllable generation to generate samples conditioned on the observation $\mathbf{y}$. We note that the direct sampling from the pretrained score can only yield T snapshots $\mathbf{X} = \{\mathbf{x}^t\}_{t=1}^T$ (that is, the number of generated snapshots is limited by the chosen T), while the observation may depend on a long sequence $\mathbf{y} = \mathbf{H}(\{\mathbf{x}^t\}_{t=1}^T) + \boldsymbol{\epsilon}$ with $T \gg T$ as the total number of snapshots. To generate predictions over longer sequences, we divide the sequence into two parts by distinguishing them as observation-dependent or independent subsequences (Extended Data Fig. 2a).

For the subsequence depending on the observation $\mathbf{y}$, we first initialize B consecutive samples $\mathbf{X}_{t=1}^{(i)} = \{\mathbf{x}_{t=1}^{(i)}\}_{t=1}^{i-T-(i-1)-m} \sim \mathcal{N}(\mathbf{0}, \sigma_{t=1}^2 \mathbf{I}_d)$, $i = 1, \dots, B$, where m is the number of snapshots overlapping with the previous sample and B the total number of decomposed pieces (for example, $B = 4$ and $m = 2$ in Extended Data Fig. 2b). Here, we assume $\sigma_{t=1}$ is sufficiently large, and then obtain $\mathbf{X}_{t=1}^{(i)} = \mathbf{X}_{t=0}^{(i)} + \sigma_{t=1} \boldsymbol{\epsilon} \approx \sigma_{t=1} \boldsymbol{\epsilon}$. Inspired by the conditional generation model⁵⁷, we start inferring each denoised sample $\mathbf{x}_{t=0}^{(i)}$ by solving the SDE as follows:

$$\begin{aligned} \begin{bmatrix} \mathbf{X}_\tau^{(1)} \\ \mathbf{X}_\tau^{(2)} \\ \vdots \\ \mathbf{X}_\tau^{(B)} \end{bmatrix} &= -\frac{1}{2} \frac{d[\sigma_\tau^2]}{d\tau} \cdot \begin{bmatrix} \mathbf{s}_\theta(\mathbf{X}_\tau^{(1)}, \tau) \\ \mathbf{s}_\theta(\mathbf{X}_\tau^{(2)}, \tau) \\ \vdots \\ \mathbf{s}_\theta(\mathbf{X}_\tau^{(B)}, \tau) \end{bmatrix} d\tau + \sqrt{\frac{d[\sigma_\tau^2]}{d\tau}} d \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_B \end{bmatrix} \\ &\quad - \alpha_\tau \nabla_{\mathbf{X}_\tau^{(1)}, \dots, \mathbf{X}_\tau^{(B)}} \|\mathbf{y} - \mathbf{H}(\hat{\mathbf{X}}_\tau)\|_2^2 d\tau \\ &\quad - \beta_\tau \nabla_{\mathbf{X}_\tau^{(1)}, \dots, \mathbf{X}_\tau^{(B)}} \sum_{i=1}^{B-1} \left(\sum_{t=i-(T-M)+1}^{i-(T-M)+M} \|\hat{\mathbf{x}}^{t,(i+1)} - \text{sg}(\hat{\mathbf{x}}^{t,(i)})\|_2^2 \right) d\tau, \end{aligned} \quad (5)$$

where $\text{sg}(\cdot)$ is the stop-gradient operation, $\hat{\mathbf{x}}^{t,(i)} \in \hat{\mathbf{X}}_\tau^{(i)} = \mathbb{E}[\mathbf{X}_{t=0}^{(i)}|\mathbf{X}_\tau^{(i)}] = \mathbf{X}_\tau^{(i)} + \sigma_\tau^2 \mathbf{s}_\theta(\mathbf{X}_\tau^{(i)}, \tau) \approx \mathbf{X}_{t=0}^{(i)}$ (Tweedie's formula^{57,88}), $\mathbf{w}_1, \dots, \mathbf{w}_B$ are B independent standard Wiener processes and $\hat{\mathbf{X}}_\tau$ is ensembled by $\hat{\mathbf{X}}_\tau^{(1)}, \dots, \hat{\mathbf{X}}_\tau^{(B)}$ (Extended Data Fig. 2b). In equation (5), the last two terms respectively

restrict the generated samples to agree with the observation $\mathbf{y}$ (termed as observation consistency in Fig. 1d) and to share similar contents over the overlapped snapshots (termed as sequence consistency in Fig. 1d). These two penalties were balanced by two hyperparameters α_τ and β_τ (see the detailed hyperparameter tuning strategy in Supplementary Note 11 and Supplementary Table 3). We solve the SDE in equation (5) from $\tau = 1$ to $\tau = 0$ and acquire the desired denoised sequence $\mathbf{X}_{t=0}$ satisfying the observation $\mathbf{y}$.

For the subsequence not depending on the observation $\mathbf{y}$, following the denoising SDE, we first initialize a sample $\mathbf{X}_{t=1} = \{\mathbf{x}_{t=1}^t\}_{t=1}^T \sim \mathcal{N}(\mathbf{0}, \sigma_{t=1}^2 \mathbf{I}_d)$ as the initial condition for solving the denoising SDE as follows:

$$d\mathbf{X}_\tau = -\frac{1}{2} \frac{d[\sigma_\tau^2]}{d\tau} \cdot \mathbf{s}_\theta(\mathbf{X}_\tau, \tau) d\tau + \sqrt{\frac{d[\sigma_\tau^2]}{d\tau}} d\mathbf{w} - \gamma_\tau \nabla_{\mathbf{X}_\tau} \sum_{t=1}^{T_{\text{init}}} \|\hat{\mathbf{x}}^t - \mathbf{x}^t\|_2^2, \quad (6)$$

where $\hat{\mathbf{x}}^t \in \{\hat{\mathbf{x}}^t\}_{t=1}^T = \hat{\mathbf{X}}_\tau = \mathbb{E}[\mathbf{X}_{t=0}|\mathbf{X}_\tau] = \mathbf{X}_\tau + \sigma_\tau^2 \mathbf{s}_\theta(\mathbf{X}_\tau, \tau) \approx \mathbf{X}_{t=0}$ (Tweedie's formula), and $\{\mathbf{x}^t\}_{t=1}^{T_{\text{init}}}$ are the given initial snapshots ($T_{\text{init}} < T$, $T_{\text{init}} = 2$ in Extended Data Fig. 2c, for example). The last term on the right-hand side of the above SDE constrains the generated samples to satisfy the given initial snapshots, and γ_τ is the hyperparameter to balance the effect of this term (see the detailed hyperparameter tuning strategy in Supplementary Note 11 and Supplementary Table 3). By solving the SDE in equation (6) from $\tau = 1$ to $\tau = 0$, we can acquire a sample containing the prediction of the subsequent snapshots. In this way, a longer sequence can be generated by iteratively applying such a process in an autoregressive manner (Extended Data Fig. 2c).

Although we can change the time step size Δt during the inference (generating) stage, the model performs best using a consistent time step size used for training and testing (see Supplementary Note 10 and Supplementary Figs. 27 and 28 for details).

Evaluation metrics

We choose nRMSE⁹ as the primary evaluation metric, defined as

$$\text{nRMSE}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{\|\mathbf{x} - \hat{\mathbf{x}}\|_2}{\|\mathbf{x}\|_2},$$

where $\mathbf{x}$ is the ground-truth data and $\hat{\mathbf{x}}$ is the predicted data. In this context, a lower nRMSE indicates a higher pointwise accuracy. We also use the Pearson correlation coefficient in the ERA5 test to show the correlation between the reference and the predictions, defined as

$$\text{Correlation}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{(\mathbf{x} - \mathbf{x}_{\text{mean}}) \cdot (\hat{\mathbf{x}} - \hat{\mathbf{x}}_{\text{mean}})}{\|\mathbf{x} - \mathbf{x}_{\text{mean}}\|_2 \cdot \|\hat{\mathbf{x}} - \hat{\mathbf{x}}_{\text{mean}}\|_2},$$

where $\mathbf{x}, \hat{\mathbf{x}}$ are two vectors with the same dimension, and the subscript 'mean' refers to the mean value of all the entries of the vector. The Pearson correlation coefficient is a value between -1 and 1 , and a higher value indicates a stronger correlation between these two vectors.

For the relative error field reported in Fig. 2b,e,h, each entry of the error field is defined as

$$r_i = \frac{|x_i - \hat{x}_i|}{|x_i|}$$

where x_i and $\hat{x}_i$ represent the i th entries of the ground-truth data $\mathbf{x}$ and predicted data $\hat{\mathbf{x}}$, respectively, and $|\cdot|$ denotes the absolute value of the input.

We also define the statistical feature to quantify the quality of prediction for chaotic dynamics, on the basis of the spatial two-point correlation defined as⁷²

$$R(r) = \langle u'(x, t) u'(x + r, t) \rangle,$$

$$\text{s.t. } u'(x, t) = u(x, t) - \langle u(x, t) \rangle_x,$$

where $\langle \cdot \rangle$ denotes the average over all spatial points x and temporal points t , $\langle \cdot \rangle_x$ the average over all x , $u(x, t)$ the solution of the KSE and r the spatial lag. The spatial two-point correlation, a frequently used statistical feature to evaluate the numerical solution of the KSE⁷², measures how the values of u at two points separated by a distance r are spatially correlated. Herein, we normalize this value by $R_{\text{ref}}(r = 0)$, which is the two-point correlation of the reference solution at $r = 0$.

The statistical feature of the Kolmogorov turbulent flow is defined in the context of the kinetic energy spectrum (usually used in the analysis of turbulent systems), expressed as⁶⁹

$$E(\lambda) = \pi \lambda \langle |\mathbf{u}(\lambda)|^2 \rangle.$$

Here, $\langle \cdot \rangle$ denotes the average over all λ in the Fourier space, and λ the amplitude of the $\mathbf{u}$, that is, $\lambda = |\mathbf{u}|$.

Experimental set-up for cylinder flow measurements

We conducted a laboratory experiment on the PIV measurements (Extended Data Fig. 3) of the cylinder flow under different Reynolds numbers ranging from 100 to 250. The experimental instruments and set-ups are inspired by an earlier investigation of cylinder flow⁸⁹. The reason for not choosing higher or lower Reynolds numbers is that with a fixed diameter of the cylinder a lower Reynolds number corresponds to lower moving speed, resulting in markedly enhanced background low-frequency noise in our experiment. The observing window for the PIV measurements is chosen from the close downstream of the cylinder. The measured quantities with dimensions are non-dimensionalized using the cylinder's diameter and movement speed to facilitate further processing. Since the PIV technique allows us to obtain relatively complete information about the velocity field, the data for prediction are degenerated from the PIV measurements.

Baseline methods

We consider seven different baseline models, namely, U-Net^{73,90}, FNO³⁹, DeepONets⁷⁵, LNO⁷⁴, PINN^{43,44}, DMD¹⁶ and piDMD³⁴.

U-Net is a convolution-based architecture containing a series of downsampling and upsampling modules. Despite being introduced for image segmentation⁷³, U-Net can be used for reconstruction of spatiotemporal dynamics by adopting the observation data as its inputs while making predictions of the full field^{8,90}. In our experiments for reconstruction, the input of the U-Net consists of ten snapshots of the observations (which are interpolated to a regular grid to build an image-like data structure), while the output includes ten reconstructed snapshots. In this way, the U-Net architecture can share the same training pipeline as the other operator-based NNs (see Supplementary Note 2 for detailed set-ups). FNO attempts to map between function spaces, leveraging the global information of Fourier features³⁹. DeepONets leverage the universal operator approximation theorem to design a network architecture that can learn the mapping between function spaces⁷⁵, while they may not perform well in systems with very complicated spatiotemporal patterns due to the limited model capacity. LNO further incorporates a Laplace layer that can better serve for modelling specific dynamical systems⁷⁴ (for example, systems with non-periodic signal and transient response). These neural methods are compared in the KSE and Kolmogorov flow for both reconstruction and forecasting tasks (Figs. 2 and 3), trained in an end-to-end manner that adopts the observation data as input while predicting the complete data (see Supplementary Note 2 for detailed experimental set-ups). PINN is a famous deep learning paradigm that integrates the residual of physical laws into the loss function as a regularizer or soft constraint. By minimizing such a regularized loss, it can handle many forward and inverse analyses of partial differential equation systems. Parameter settings of these NN-based methods (that is, FNO, U-Net, DeepONets, LNO and PINN) are listed in Supplementary Table 2.

As a conventional approach, DMD is a linear data-driven method that has been widely used for analysing dynamical systems via approximation of dynamics with a learned linear operator¹⁶. By constraining the linear operator on a specific manifold, piDMD injects our prior physics knowledge into the DMD model to make it more robust and accurate³⁴. The detailed experimental set-ups for these methods can be found in Supplementary Note 7.

Data availability

All datasets used in this work are available online. The datasets for the KSE and Kolmogorov flow are available via the Zenodo repository⁹¹ at <https://doi.org/10.5281/zenodo.13925732>. The ERA5 reanalysis data⁷¹ are available at <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=download>. The datasets for the compressible Navier–Stokes equation and Burgers' equation can be found in the GitHub repository of PDEBench⁹⁰ at <https://github.com/pdebench/PDEBench> (see Supplementary Note 3 for details). Source data are provided with this paper.

Code availability

The source codes to reproduce the results in this study are available via a Code Ocean capsule at <https://doi.org/10.24433/CO.6670426.v2> (ref. 92).

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Author contributions

L.Y., Y.D., W.H. and H.S. supervised the project. L.Y., Z.L. and W.H. conceived the idea. Z.L. carried out the numerical simulations. Z.L. and Y.Z. designed and carried out the experiments. Z.L. and W.H. performed the machine learning studies. Q.F., J.L., L.Q. and R.D. discussed the machine learning results. All authors discussed the results and assisted during manuscript preparation.

Competing interests

The authors declare no competing interests.

Additional information

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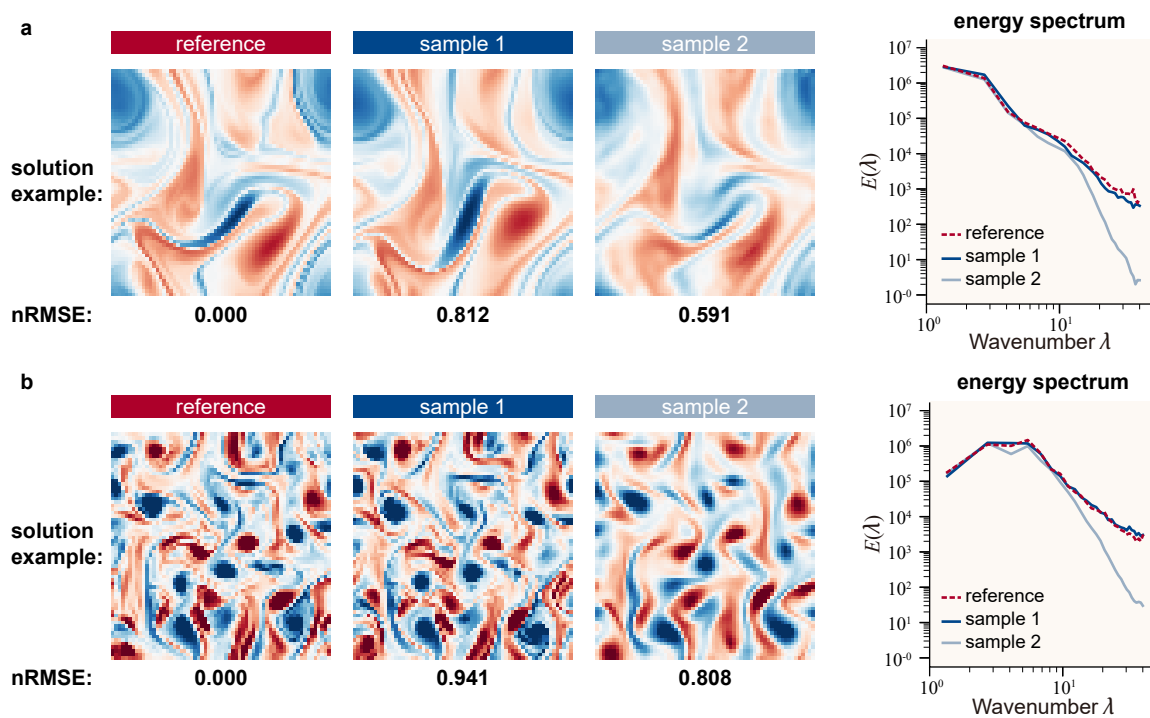
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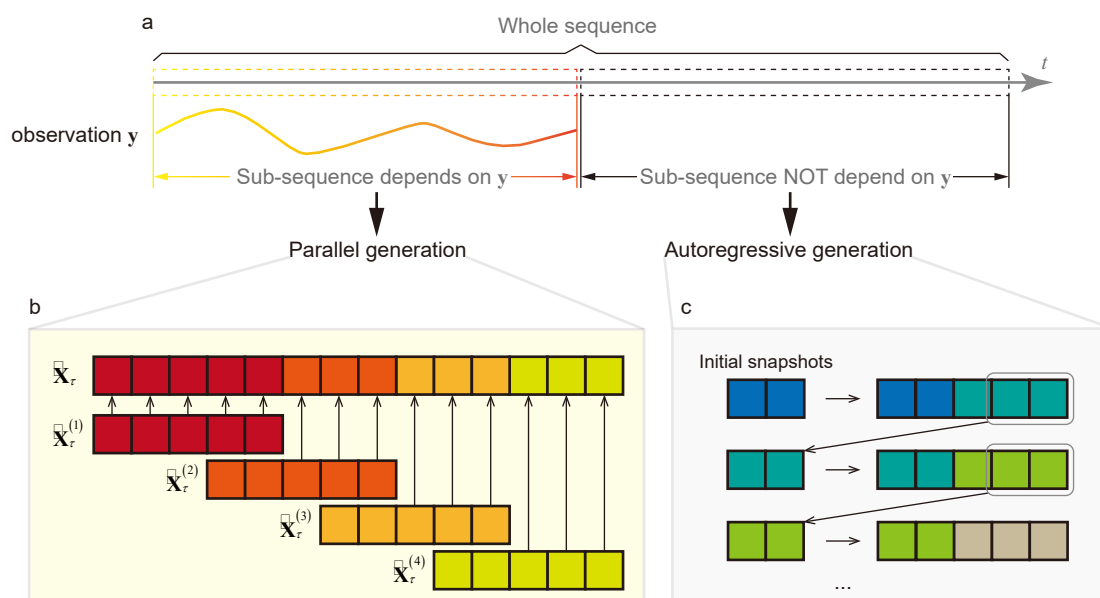
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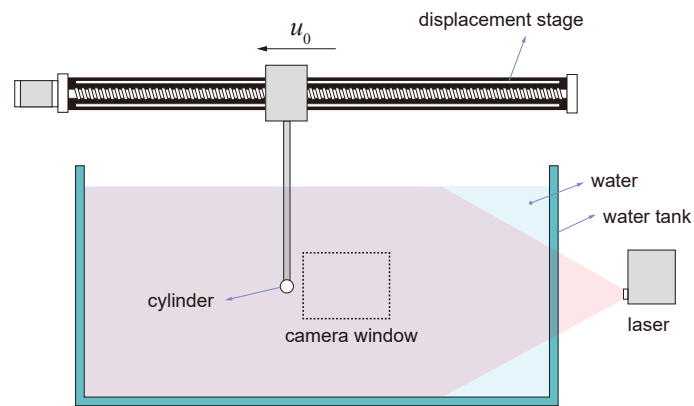
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Extended Data Fig. 1 | Demonstration of pointwise accuracy and spectral accuracy. a, Two samples are compared to the reference, in which sample 2 shows a lower pointwise error while failing in capturing the spectral feature. **b**, Another case at different parameters that shows the identical problem as in **a**.



Extended Data Fig. 2 | Strategy for generating a longer sequence. a, The entire sequence is divided into two subsequences according to their dependency on observation y . **b,** Parallel generation for subsequence depending on observation y . **c,** Autoregressive generation for subsequence not depending on observation y .



Extended Data Fig. 3 | Schematic of PIV experimental setup. The velocity measurement of cylinder flow is conducted using PIV technique. We use a motorized displacement stage to drive both the circular cylinder and video recorder. The aluminum cylinder with diameter of 6 mm and length of 38 mm is suspended in a glass tank filled with water, while the video recorder is also

attached to the displacement stage to move synchronously with the cylinder. A beam of sectorial helium-neon laser irradiates the cylinder vertically from the downstream. To avoid severe reflection of laser, the cylinder has a matte finish all over its surface.